Subhadeep Pal

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Cambridge, Massachusetts, United States

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Employment

- **Massachusetts Institute of Technology, Cambridge, MA** *Jan, 2026 - Present*
  PostDoc in Dept. Civil & Environmental Engineering
  Advisor: *Markus Buehler*

Education

- **Northwestern University, Evanston, IL** *Sep, 2020 – Dec, 2025*
  PhD in Mechanics, Materials, and Structures, Dept. of Civil & Environmental Engineering GPA: 3.93/4
  Committee: *Sinan Keten (Chair), Wei Chen, Mitchell Wang*
- **Indian Institute of Technology, Kharagpur, India** *Aug, 2015 – May, 2020*
  BTech and MTech (integrated) in Civil & Structural Engineering (Rank: 2/84) GPA: 9.22/10

Research Interest

Computational soft materials; Polymer physics; Extreme Mechanics, Polymer-grafted nanoparticles; Crosslinked polymer networks; Coarse-graining strategies; Multiscale modeling; Molecular simulations; Mechanics of soft matter; AI- & data-driven materials design; Large Language Models, Cementitious Materials; Finite Element Analysis.

Honors and Awards

- **Padden Award Symposium Finalist, Division of Polymer Physics (DPOLY)** *2025*
  Chosen as a finalist to present my research at the prestigious Frank J. Padden, Jr. Award Symposium at the APS Global Physics Summit, a distinction awarded to only **5-10 Ph.D. students** globally through a rigorous selection process.
- **Ryan Fellowship, International Institute of Nanotechnology, NU** *2023-2025*
  Awarded in recognition of excellence in research, and selected as the **1st recipient** from the Dept. of CEE. This fellowship, one of the university's most prestigious, is awarded to only **14 recipients** across all Ph.D. programs.
- **NSF ACCESS Grant, NU** *2023-2025*
  Secured **400,000** compute hours through a highly selective process to support my research. Wrote a successful **research grant** and acted as the Computational Investigator (CI) on the grant.
- **NSF Student Travel Award, SES** *2025*
  Received **$800** travel award from NSF Mechanics of Materials and Structures (MOMS) to attend the Society of Engineering Science conference. **50 people** are chosen out of **299 applicants**
- **Student Travel Award, USNCCM18** *2025*
  Received **$430** travel award to attend 18th US National Congress Computational Mechanics conference.
- **Article on Spotlight, The Graduate School, NU** *2024*
  Featured among Northwestern's most distinguished ∼**3,500 PhD students**.
- **Murphy Fellowship, The Graduate School, NU** *2020-2021*
  Received this prestigious fellowship to fund the 1st year of PhD study based on academic merit.
- **UTSIP Scholarship, The Graduate School of Frontier Science, UTokyo** *2019*
  Awarded the research fellowship from the University of Tokyo through an intensely competitive selection process, with only **15 individuals** chosen from a global pool of applicants.

Publications [Citation: 127, h-index: 7, i10-index: 7]

16. Wang, F. Y., Marom, L., **Pal, S.**, Luu, R. K., Lu, W., Berkovich, J. A., & Buehler, M. J. (**2026**) Autonomous Agents Coordinating Distributed Discovery Through Emergent Artifact Exchange. https://arxiv.org/abs/2603.14312 (*arXiv preprint*)
15. Singh, M., Lin, H., **Pal, S.**, Zhou, C., Mohammed, M., Maddukuri, S., Deen, P., Yin, R., Bockstaller, M., Matyjaszewski, K., Smith, Z., Karim, A., Kysar, J., Keten, S., Benicewicz, B., & Kumar, S. K. (**2026**) Process-Driven Control of the Properties of Polymer Grafted Nanoparticle Composites. ***ACS Nano*** (*In press*)
14. **Pal, S.**, Wu, Z., Joodaki, F., Cusatis, G., & Keten, S. (**2026**) Molecular Investigation of Freshly-made Cement Paste Rheology for 3D-printing Application. ***Journal of Physical Chemistry*** (*In review*)
13. **Pal, S.**, Khumbhojkar, G., Chen, W., & Keten, S. (**2026**) From Packing to Performance: Decoding Nanoscale Mechanics of Binary Polymer-Grafted Nanoparticles via Active Learning. ***Macromolecules*** (*In press*)
12. Moussavi, A., Wu, Z., **Pal, S.**, & Keten, S. (**2025**) Nanoconfinement Release Toughens Polymer Grafted Nanoparticles Assemblies Through Better Interdigitation and Entanglements. ***ACS Nano Letters*** *25*(16), 6637–6644 (DOI: 10.1021/acs.nanolett.5c00680)

11. Xiao, R., **Pal, S.**, Rademacher, C., Chen, J., Wang, Q., Chen, W., Keten, S., Shull, K. R., & Wang, M. (**2025**) Real-Time Visualization of Polymer Single Chain Conformational Change in Bulk State Using Super-Resolution Microscopy. ***Physical Review Letters***, *134*(14), 148101 (DOI: 10.1103/PhysRevLett.134.148101) [**Editor's Suggestion**]
10. Chen, T., Qian, Y., Laine, A., Kwon, J., Liu, L., **Pal, S.**, Gupta, S., Vargo, E., Su, G. M., Ritchie, R. O., Keten, S., Wang, R., Salmeron, M., & Xu, T. (**2025**). Engineer Reversible Nanocomposite by Programming Amorphous Polymer Conformation Under Nanoconfinement. ***Advanced Materials***, 2415352 (DOI: 10.1002/adma.202415352)
9. Hafner, B., **Pal, S.**, Lewis, B., Keten, S., & Shull, K. (**2024**). Network Topology and Percolation in Model Covalent Adaptable Networks. ***ACS Macro Letters***, *13*, 1545–1550 (DOI: 10.1021/acsmacrolett.4c00523) [Journal Cover]
8. **Pal, S.**, & Keten, S. (**2024**). Micro-Ballistic Response of Thin Film Polymer Grafted Nanoparticle Monolayers. ***Soft Matter***, *20*(39), 7926–7935 (DOI: 10.1039/D4SM00718B) [**Soft Matter Pioneering Investigators series**]
7. Moussavi, A., **Pal, S.**, Wu, Z., & Keten, S. (**2024**). Characterizing the Shear Response of Polymer-Grafted Nanoparticles. ***Journal of Chemical Physics***, *160*(13) (DOI: 10.1063/5.0188494) [**Special Collection: Polymer Nanoconfinement**]
6. **Pal, S.**, Dansuk, K. C., Giuntoli, A., Sirk, T., & Keten, S. (**2023**). Predictive Molecular Modeling of Hardener Composition Effects on Epoxy Resin Fracture Properties. ***Macromolecules***, *56*(12), 4447–4456 (DOI: 10.1021/acs.macromol.2c02577)
5. Dansuk, K. C., **Pal, S.**, & Keten, S. (**2023**). A Catch Bond Mechanism with Looped Adhesive Tethers for Self-Strengthening Materials. ***Nature Communications Materials***, *4*(1), 60 (DOI: 10.1038/s43246-023-00387-y)
4. Wu, Z., **Pal, S.**, & Keten, S. (**2023**). Implicit Chain Particle Model for Polymer-Grafted Nanoparticles. ***Macromolecules***, *56*(9), 3259–3271 (DOI: 10.1021/acs.macromol.2c02272)
3. **Pal, S.**, & Mitra, N. (**2021**). Shock wave propagation through air: a reactive molecular dynamics study. ***Proceedings of the Royal Society A***, *477*(2247), 20200676 (DOI: 10.1098/rspa.2020.0676)
2. **Pal, S.**, Mitra, N., Sarkar, P. K., & Prasad, D. (**2020**). Stretch-induced helix to extended coil transition of crystalline α phase isotactic polypropylene: A molecular dynamics study. ***Polymer Crystallization***, *3*(4), e10143 (DOI: 10.1002/pcr2.10143) [Journal Cover]
1. Karthika, P. S., Verma, A., & **Pal, S.** (**2020**). Integration of image processing and crowd simulation for understanding crowd dynamics in Kumbh Mela. ***Transportation Research Procedia***, *48*, 2906–2915 (DOI: 10.1016/j.trpro.2020.08.229)

ORAL PRESENTATIONS

- *20th U.S. National Congress on Theoretical and Applied Mechanics, Pasadena, California* 2026
- *Society of Engineering Science, Atlanta, Georgia* 2025
- *18th U.S. National Congress on Computational Mechanics, Chicago, Illinois* 2025
- *Engineering Mechanics Institute Conference, Anaheim, California* 2025
- *Frank J. Padden, Jr. Award Symposium, APS March Meeting, Anaheim, California* 2025
- *Ryan Fellows Meeting, Evanston, Illinois* 2024
- *APS March Meeting, Minneapolis, Minnesota* 2024
- *SES Annual Technical Meeting, Minneapolis, Minnesota* 2023
- *CHiMaD Annual Meeting, Chicago, Illinois* 2023
- *APS March Meeting, Chicago, Illinois* 2022
- *APS March Meeting, Denver, Colorado* 2020

RESEARCH PROJECTS

- **Structured Scientific Reasoning in Open-Weight LLMs via Reinforcement Learning** *2026 - Present*
  Dr. Markus Buehler
  - Fine-tuned open-weight LLMs (Qwen3-8B) via **Supervised Fine-Tuning** and **Group Relative Policy Optimization (GRPO)** to instill structured reasoning outputs, including chain-of-thought and knowledge graph formats with domain-specific tags (e.g., *<brainstorm>*, *<graph>*, *<patterns>*).
  - Improved reasoning quality on **hypothesis generation** tasks by ∼**15%** across scientific benchmarks (**MMLU-Pro**, **ChemBench**, **FrontierScience**) relative to the base model.

- **Decentralized Multi-Agent AI for Scientific Discovery** *2026 - Present*
  Dr. Markus Buehler, Fiona Wang, Dr. Anubhav Jain (LBNL)
  - Co-developed **ScienceClaw + Infinite**, a decentralized multi-agent framework in which autonomous agents **chain 300+ scientific skills** to drive open-ended materials discovery without a central coordinator or human-in-the-loop.
  - Engineered the **polymer-physics skill module** and a multi-agent elastomer design workflow within ScienceClaw, integrating chain-statistics models and MD-derived surrogates to autonomously generate hypotheses and **computationally validate** design candidates in a shared artifact ecosystem.

- **Micro-Ballistic Performance of PGNs Thin Film Monolayers** *2022 - 2024*
  Dr. Sinan Keten
  - Performed the micro-ballistic **LIPIT** on coarse-grained SiO_2-PMMA polymer-grafted nanoparticles (PGNs) thin films via molecular simulations to examine the effect of grafted **chain length** and **nanoparticle (NP) core densities** on the ballistic performance, as PGNs exhibit **remarkable mechanical properties**.
  - Identified the positive relationship between **cohesive energy density** and **specific penetration energy (E_p^*)** as well as the change in failure method from **segment pull-out** to **fragmentation** in shorter grafted chains due to **low cohesion**.
  - Formulated a new ballistic metric called average **deceleration rate of bullet (A_b)**, which measures the state of the bullet after it passed through thin film and realized intermediate grafts (N = 40-100) provide **optimal** E_p^*, and A_b, promising them for **impact resistant** applications where backface deformation is undesirable.

- **Investigating Mechanical Properties of Multi-Component Polymer Grafted Nanoparticles** *2023 - 2025*
  Dr. Sinan Keten, Dr. Wei Chen
  - Constructed **molecular models** of binary PGNs with different graft architectures, reproducing experimentally observed PGN superlattices and elucidating their **mechanical responses**. The model revealed that ***NaCl***-like binary PGNs offer both higher modulus and toughness **simultaneously** due to efficient packing.
  - Implemented the **Gaussian process** (GP) metamodel and **Bayesian optimization** (BO) to complement CG-MD results and efficiently probe the complex design space. This framework reduces uncertainty, constructs the **Pareto front**, and develops **structure-property mapping**.

- **Predicting Mechanical and Fracture Properties of Mixed Hardener Epoxies** *2022 - 2023*
  Dr. Sinan Keten, Dr. Timothy Sirk (ARL), Dr. Kerim Dansuk (Bogazici Uni.), Dr. Andrea Giuntoli (U. of Groningen)
  - Utilized a **chemistry-specific** coarse-grained model to investigate the impact of **mixed-hardeners** on the mechanical properties of epoxy resins, providing insights regarding the influence of hardener compositions on the **failure mechanisms** and the strength-toughness trade-off.
  - Estimated **fracture toughness** based on a continuum-based fracture model using simulation results that accurately captured **experimental trends**.
  - Demonstrated that the **intermediate-length** Jeffamine chains enhance toughness while maintaining moduli, which established the **predictive framework** for resin design, overcoming the challenges of the brittleness in epoxy.

- **Characterizing the Shear Response of PGNs** *2022 - 2023*
  Dr. Sinan Keten, Arman Moussavi, Dr. Zhenghao Wu (Xi'an Jiaotong Uni.)
  - Simulated PGN with **Kremer-Grest** beads and **FENE** bonds to evaluate shear response with varying nanoparticle core radius, grafting density, and chain length, which demonstrated a **linear** relation between **NP volume fraction** and **modulus**.
  - Discovered PGNs with short grafted chains has **higher stiffness** than linear polymer melts due to **reinforcement** from NP, suggesting the potential for creating **stronger polymer composites**, a key advancement critically needed to **reduce** the reliance on **single-use plastics**.

- **A Catch Bond Mechanism with Looped Adhesive Tethers** *2021 - 2023*
  Dr. Sinan Keten, Dr. Kerim Dansuk (Bogazici Uni.)
  - Developed a **molecular design** for NP-based system mimicking **catch bond** characteristics using looped and straight tethers, enabling load-sharing under mechanical force, which demonstrated a **force-enhanced** lifetime curve in simulations.
  - Derived an **analytical relation** validated by molecular and Monte-Carlo simulations to tailor loop and adhesion interactions, achieving **customizable** catch bond lifetime curves and advancing the design of **tunable, self-strengthening** materials.

- **Development of a Particle-based Model for PGNs** *2022 - 2024*
  Dr. Sinan Keten, Dr. Wei Chen, Dr. Sanat K Kumar (Columbia), Dr. Zhenghao Wu (Xi'an Jiaotong Uni.)
  - Designed an implicit chain particle model (ICPM) framework to represent the PGNs with a **single bead**, achieving $\sim 10^7$ times **computational speed-up** compared to all-atomistic models and **accurately predict** the bulk modulus and toughness.
  - Reproduced uniaxial tensile response at small strains and captured **rate-dependent** mechanical behavior with ICPM, validating its application for mesoscale PGN mechanics.

- **Understanding Cement Paste Rheology for 3D Printing Application** *2023 - 2025*
  Dr. Sinan Keten, Dr. Faramaz Joodaki, Dr. Gianluca Cusatis

- Create a **nanoscale** model for freshly-made cement paste using C3S & water and compute the **potential of mean force** (PMF) between two cement surfaces at varying water content between them to develop a micron-scale cement model.
- Reproduce the cement flow through the **3D-printer nozzle** with the micron scale model and discover the **viscosity** of cement paste reduces with increasing flow (shear) rates and higher water-cement ratios.

- **Real-Time Super-Resolution Imaging of Polymer Chain Conformational Changes** *2023 - 2024*
  Dr. Sinan Keten, Ruiqi Xiao, Dr. Muzhou Wang
  - Formulated **finite element models** of bottlebrush polymer thin films with a rigid indenter and a friction contact between them to simulate indentation experiments with a 10^7 dimension scaling in **ABAQUS**.
  - Analyzed the **orientation change** of the individual polymers by defining uniformly distributed vectors and mapping their displacement using the simulated **strain field**. The simulations validated the experimental observations, which offered insights into the movement of **individual polymers**, an innovative technique in the polymer physics field.

- **Engineering the Mechanics of Nanocomposites via Polymer Conformation Programming** *2023 - 2024*
  Dr. Sinan Keten, Tiffany Chen, Dr. Ting Xu (UC Berkeley)
  grad
  - Employed the **ICPM** framework to construct micron-scale PGN thin films with nanoparticle radii 10–100 $\times$ larger than graft lengths, enabling efficient simulation of **nanoindentation** responses and quantitative **modulus comparison**.
  - Achieved close agreement with experimental **modulus trends** and observed failure patterns, revealing that stiffness enhancement arises from **increased inter-PGN cohesion** mediated by **activated polymer volume**; a molecular-level mechanism inaccessible to experiments alone.

- **Network Topology and Percolation in Model Covalent Adaptable Networks** *2023 - 2024*
  Dr. Sinan Keten, Benjamin Hafner, Dr. Kenneth Shull
  - Developed coarse-grained molecular models of **recyclable disulfide-based** epoxies with LAMMPS and in-house MATLAB codes to compliment experimental **network analysis** study.
  - Explored the **complex design space**, which is challenging and resource-intensive to examine experimentally, to identify the **percolation threshold**, finding strong alignment with Flory's simplified **mean-field theory**.

Teaching Experiences

- **Teaching Assistant & Grader in Graduate-level course *Finite Element Methods in Mechanics*** *Fall 2022*
  Coordinated work among teaching assistants, evaluated assignments, and guided 36 students during office hours.
- **Grader in Graduate-level course *Structural Analysis - Dynamics*** *Fall 2021*
  Assessed assignments, graded final examinations, and advised students on their final course projects.

Official Reviewer

Nature Chemistry, International Journal of Impact Engineering, Journal of Chemical Physics, Soft Matter, Nanoscale Advances, Journal of Physical Chemistry, npj Computational Materials, Journal of the Mechanical Behavior of Biomedical Materials

Leadership Roles

- **Treasurer & Event organizer, Graduate International Student Organization, NU** *2021 - Present*
  - Organize events for international graduate students, fostering new connections and a sense of community.
  - Manage finances, prepare annual budgets, and maintain accurate financial records.
  - Secured $16,000 in funding from the Office of Student Life for event organization.
- **Event organizer, Bengalis in Chicago, UIC** *2021 - Present*
  - Organize cultural events for *Bengalis* in the Greater Chicago area.
  - Coordinate with a team of volunteers to manage logistics for annual events, such as *Durga Puja*, with over 500 attendees.

Skills & Proficiency

- **Technical Skills:** Python, BASH, LAMMPS, ABAQUS, Ovito, MATLAB, VMD, Git, C, Materials Studio, AutoCAD.
- **Languages:** English (full proficiency), Hindi (conversational), Bengali (native).

REFERENCES

1. **Dr. Sinan Keten**
   Northwestern University
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   Relationship: Primary thesis advisor
2. **Dr. Markus Buehler**
   Massachusetts Institute of Technology
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   Relationship: Postdoctoral advisor
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